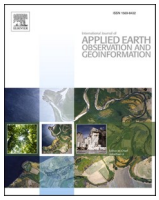
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Estimating medium-term regional monthly economic activity reductions during the COVID-19 pandemic using nighttime light data

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ABSTRACT

Economic impact estimates of the initial lockdowns due to the COVID-19 pandemic showed a significant reduction in economic activities globally. However, the succeeding impacts and their spatiotemporal distribution within countries remain unknown. Studies showed that nighttime light data (NTL) has effectively revealed the spatiotemporal dimensions of the economic effects of COVID-19. Thus, this study used NTL data to determine the medium-term regional monthly economic impacts of the pandemic in the Philippines in terms of the Economic Activity Reduction (EAR) index. We generated a spatial error model, regressing pre-pandemic NTL on mean temperature, maximum rainfall, and built-up area. This model explained 81.6% of the pre-pandemic NTL and was used to estimate the counterfactual NTL. We subtracted the actual from the counterfactual to compute the EAR. Then, the EAR was regressed on regional factors to determine which ones influence the impacts. Results showed uneven distribution of EAR across space and time. The EAR was generally higher in urban regions than in rural ones. Overall, more regions in the south had higher EAR. Temporally, the EAR showed a dynamic pattern, increasing in less urban regions and decreasing in highly urbanized regions. Regional analysis showed that urbanization level, population density, and poverty incidence had a significant positive relationship with the EAR. Beyond the immediate impacts, NTL effectively revealed spatiotemporal dimensions of the economic effects of a long-term global hazard.

1. Introduction

The coronavirus disease 2019 (COVID-19) spread rapidly across countries and contracted the global economy by approximately −3.4 % in 2020 (World Bank Group, 2022). Immediate macroeconomic losses due to COVID-19 were estimated for various countries using conventional methods (Kim et al. 2022; Betho et al. 2021; Santos et al. 2022). However, these data and methods face limitations in determining the subnational level dynamic impacts across years, delaying full recovery in local communities.

Beyond the country-level impacts, researchers pointed out the importance of subnational impact estimates for a proportionate resource distribution (Lopez Ochoa et al. 2021) and high-frequency data to monitor effects for timely policy implementation (Fezzi and Fanghella, 2020). In addressing this, a growing number of studies have been using high-resolution and high-frequency datasets to estimate near-real-time subnational impacts. For instance, researchers used electronic payments (Carvalho et al. 2021), electricity consumption (Beyer et al. 2021; Fezzi and Fanghella, 2020), mobility data (Yang et al. 2022), and night-time light (NTL) data (Alahmadi et al. 2021, 2023; Beyer et al. 2021; Xu et al. 2021) to analyze subnational high-frequency economic impacts.

Among these datasets, NTL is frequently used in determining COVID-19 impacts. NTL constitutes satellite data that records nocturnal light brightness across the earth's surface and have been used in a broad range of cases, from socioeconomic to meteorological, and environmental applications (Li et al. 2017; Miller et al. 2013). Among its applications, one critical niche for NTL is mapping economic variables, such as the GDP (Chen and Nordhaus, 2019; Zhao et al. 2017). Earlier studies (Doll et al. 2000; Henderson et al. 2012) statistically established the relationship of NTL with economic variables, paving the way for its wider acceptance as a tool for mapping economic activities. Subsequent papers supported these earlier findings using NTL for mapping spatial patterns of economic variables (Wu et al. 2013; Mellander et al. 2015).

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Recent papers capitalized on its unique capacity for detecting human activities and furthered its function to estimate losses from natural disasters and monitor recovery (Jia et al. 2023; Qiang et al. 2020). The advantages of NTL include fine spatial details, global coverage, extensive temporal record (Jia et al. 2023), free access, minimum release delay, objective physical measurement (Schippers and Botzen, 2023), and frequent and regular data acquisition.

Continuous monitoring of regional economic impacts from COVID-19 would be crucial to navigating recovery paths. COVID-19 case surges resulted in dynamic responses and impacts lasting beyond initial lockdowns. With the economic impacts predicted to last beyond this current generation (Jorda et al. 2020), region-specific measures are necessary for communities to recover. However, medium-term local dynamic impacts and the regional factors shaping them have been rarely studied as most studies focused on immediate consequences.

Hence, this study estimates the medium-term spatiotemporal distribution of economic activity reduction (EAR) during the COVID-19 pandemic in the Philippines using regional monthly NTL. Specifically, it addresses three research questions:

1) Is EAR due to COVID-19 evenly distributed across all regions?
2) If not, how had EAR temporally spread across regions?
3) What types of regional attributes influence EAR?

We propose an approach that allows subnational impact mapping and monthly monitoring covering three years. This approach entails a model estimating the pre-pandemic NTL and using this model to estimate the counterfactual NTL. The difference between this counterfactual and actual NTL gives the EAR. Then, we identified the regional factors that significantly influence regional EARs. Through this method, we can estimate the medium-term regional monthly distribution of the impacts and understand the influencing factors.

2. Literature Review

2.1. COVID-19 in the Philippines

The Philippines implemented one of the strictest and longest COVID-19 lockdowns worldwide (Hapal, 2021). This country registered the highest economic losses among Southeast Asian countries in 2020 (Zhang et al. 2022; World Bank Group, 2022). Starting March 12, 2020, Metro Manila was under a partial lockdown. By March 16, 2020, the entire Luzon Island was under Enhanced Community Quarantine, restricting gatherings and closing public transportation, schools, and non-essential businesses (Amit et al. 2021). In April 2020, over half of the population was subject to these measures (Del Castillo et al. 2024). Around mid-May 2020, the government began easing restrictions, with exceptions for high-risk areas. However, different quarantine levels were implemented across local government units as cases surged in the following months. The impacts of these case surges and interventions were unknown as most studies focused on the initial impacts.

Studies that have assessed COVID-19 economic losses in the Philippines used economic data from statistics and surveys. A linear probability model was employed to ascertain the likelihood of short-term COVID-19 business impacts based on survey data on MSMEs (Shinozaki and Rao 2021) noting greater probability of economic reduction in urban regions than in rural ones. Another study applied the persistent inoperability input–output method and identified the worst-hit sectors (Yu et al. 2020). Others derived a compartmentalized dynamic transmission model to estimate the economic and health tradeoffs for different scenarios in selected regions (de Lara-Tuprio et al. 2022). A recent study estimated a total loss amounting to 2.9 trillion PhP in 2020 and 3.2 trillion PhP in 2021 due to COVID-19 in the Philippines (Del Castillo et al. 2024). However, the regional monthly economic impacts of COVID-19 across time remain unknown.

2.2. COVID-19 economic impact estimation methods

Internationally, COVID-19 impacts have been estimated in various countries using traditional methods such as the social accounting matrix for Mozambique (Betho et al. 2021), computable general equilibrium in South Korea (Kim et al. 2022), and input–output models in the Philippines (Santos et al. 2022). With the need for near-real-time impact estimation, other researchers used high-frequency datasets. Carvalho et al. (2021) used electronic payment data to analyze the regional economic impacts of COVID-19 in Portugal, finding central and densely populated municipalities suffering a worse economic decline. Fezzi and Fanghella (2020) used daily electricity consumption data to estimate the immediate daily economic impacts of COVID-19 in Italy. Yang et al. (2022) used daily mobility data for losses and recovery rate assessment during the initial lockdown for cities in China.

Other studies revealed the spatial distribution of impacts using NTL, as an established proxy for economic activities given the lack of local high-frequency economic data. Most NTL-based studies employed a before-and-after comparison method (Ghosh et al. 2020; Beyer et al. 2021; Alahmadi et al. 2021, 2023; Xu et al. 2021). Alahmadi et al. (2021) observed pandemic-related NTL brightness reduction in parts of Saudi Arabia after comparing 2019 and 2020 NTL images. Beyer et al. (2021) mapped out the impacts of COVID-19 lockdowns per district in India from April to August 2020 using VIIRS data. Xu et al. (2021) compared spatial and temporal changes of NTL brightness in 20 selected megacities using NASA's Black Marble NTL, identifying Mumbai, Dhaka, and Manila as the megacities with the highest nighttime brightness reduction. Other studies developed regression models (Roberts, 2021; Lin and Rybnikova, 2023) and used machine learning methods (Dasgupta, 2022). Lin and Rybnikova (2023) provided evidence that NTL can be used in estimating GDP under the impact of COVID-19 and showed the different recovery rates across U.S. states. While numerous studies have proven the utility of NTL data and other high-resolution/high-frequency datasets, most of them focused on the immediate effects covering the first few months after the lockdown.

3. Materials and methods

3.1. Study Site

The Philippines is an archipelagic country in South East Asia (Fig. 1). It has three main island groups: Luzon, Visayas, and Mindanao, and four administrative division levels: regional, provincial, municipal or city, and barangay. These administrative divisions demarcate resources, services, and policies. National services are organized in the regional units, while the barangays govern local communities.

During the COVID-19 pandemic, the National Government initially enforced lockdowns. Later, each local administrative unit implemented additional measures. These units vary in demographic, socioeconomic, and geopolitical characteristics, which could affect how the economic impacts of the pandemic spread. Understanding these differences is essential for developing region-specific recovery strategies. One key factor influencing economic impact is urbanization level, defined as the proportion of the urban population relative to the total population within an administrative unit (Philippine Statistics Authority, 2020).

In this study, we generated the model using provincial-level datasets to meet the minimum data point requirements for the spatial regression method. We aggregated the estimated provincial impacts at the regional level for interpretability. We ranked and labeled the regions according to their urbanization level, with R1 representing the most urbanized and R17 the least urbanized. This ranking helps highlight differences in urban-rural regional impacts and simplifies interpretation.

3.2. Data

Table 1 lists the data inputs used in the study with their

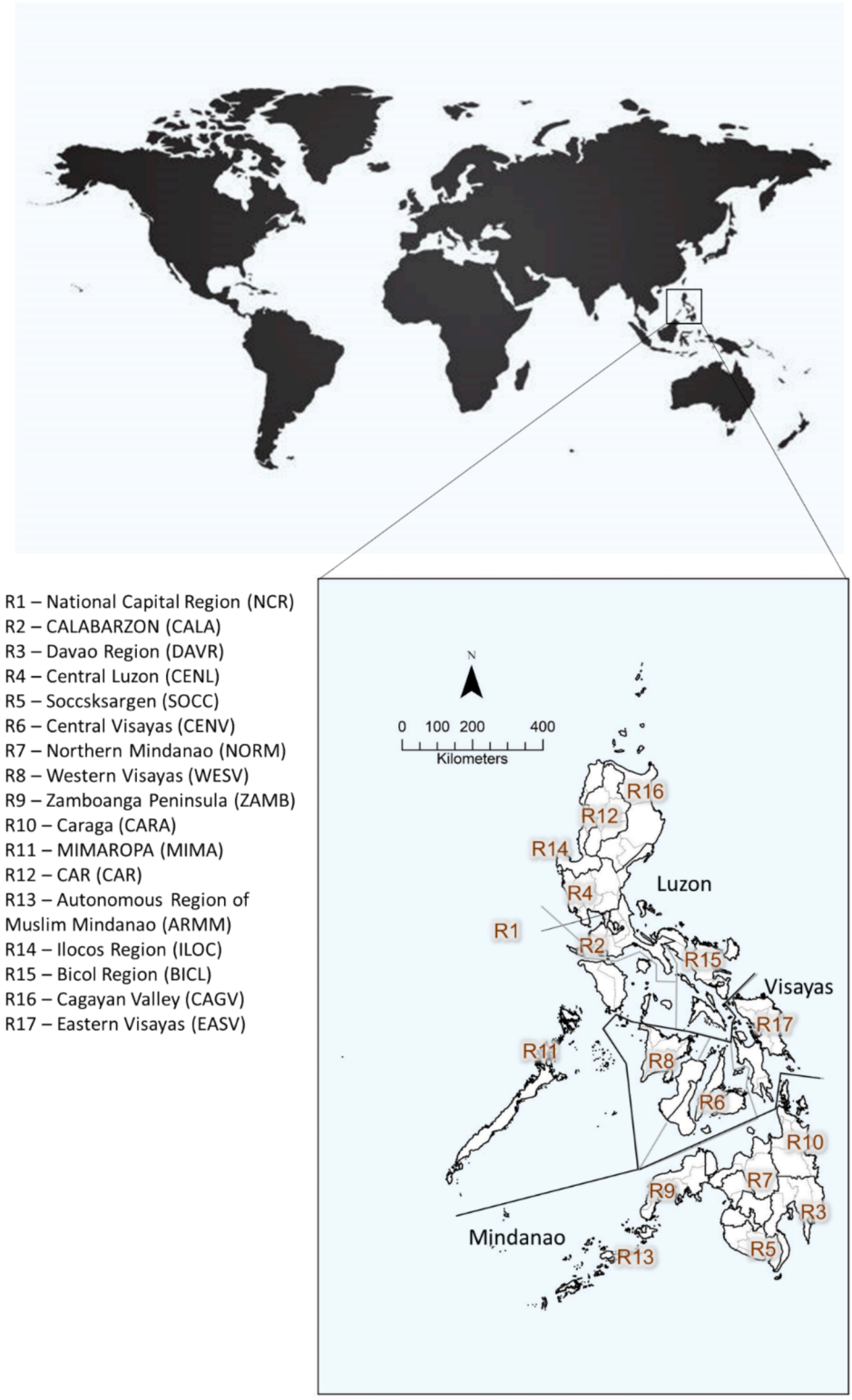


Fig. 1. Location of the Philippines and its regions ranked according to urbanization level. Note: Regional names are also provided together with abbreviations.

corresponding sources, spatial and temporal resolutions and level of aggregation. Each dataset is explained in the subsequent subsections. Table 2 presents the statistical summary of the data inputs.

3.2.1. *Nighttime light*

Monthly and annual NTL composites were downloaded from the Earth Observation Group of the Payne Institute for Public Policy, Colorado School of Mines (Earth Observation Group 2021, Elvidge et al. 2013). The Earth Observation Group preprocesses NTL data from the Suomi National Polar-orbiting Partnership, Visible and Infrared Imaging Suite Day and Night Band to eliminate unnecessary light emissions (Elvidge et al. 2017). Monthly and annual radiance average bands were then produced from the preprocessed datasets (Elvidge et al. 2021). Additional preprocessing steps via filtering and substitution were

Table 1
List of data.

Symbol	Variable Description (years covered)	Data Source	Raw Spatial resolution	Raw Temporal resolution	Level of aggregation
NTL	Nighttime light (2013–2022)	Suomi-NPP VIIRS DNB band from Earth Observation Group, Payne Institute for Public Policy, Colorado School of Mines https://eogdata.mines.edu/nighttime_light/monthly/v10/	~500 m	Monthly	Average monthly provincial
meantemp	Average daytime land surface temperature (2013–2022)	MOD11A1 V6.1 from NASA LP DAAC, USGS EROS Center https://lpdaac.usgs.gov/products/mod11a1v061/	~1000 m	Daily	Average monthly provincial
maxrainfall	Daily total rainfall (2013–2022)	Climate Hazards Center Infrared Precipitation with Station Data (Version 2.0 Final) from Climate Hazards Center, UC Santa Barbara https://developers.google.com/earth-engine/datasets/catalog/UCSB-CHG_CHIRPS_DAILY#description	5566 m	Daily	Maximum total monthly provincial
builtarea	Land cover type (2013–2022)	MCD12Q1.061 MODIS Land Cover Type Yearly Global from NASA LP DAAC at USGS EROS Center https://lpdaac.usgs.gov/products/mcd12q1v061/	500 m	Annual	Provincial built-up area
caseperH	COVID-19 case per hospital (2020–2022)	Department of Health https://data.gov.ph/index/public/dataset/covid-19-doh-data-drop-(january-01,-2022)/n9j4ekug-owr5-qmhx-6dot-4fe99tdkuw7z https://nhfr.doh.gov.ph/StatHfRegionalList	City/Region	Daily	Regional monthly
LI	Lockdown index (2020–2022)	Inter-Agency Task Forces for the Management of Emerging Infectious Diseases Resolutions	Province	~ Monthly	Regional monthly
urblev	Urbanization level (2020)	2020 Census of Population and Housing (PSA, 2022b) https://psa.gov.ph/content/urban-population-philippines-2020-census-population-and-housing	Region	Annual	Regional Annual
popden	Population density (2020)	2020 Census of Population and Housing (PSA, 2021) https://psa.gov.ph/content/2020-census-population-and-housing-2020-cph-population-counts-declared-official-president	Region	5 years	Regional 5 years
popgr	Population growth rate (2020)	2020 Census of Population and Housing (PSA, 2021) https://psa.gov.ph/content/2020-census-population-and-housing-2020-cph-population-counts-declared-official-president	Region	5 years	Regional 5 years
regmov	Percent regional movers (2020)	2020 Census of Population and Housing (PSA, 2023b) https://psa.gov.ph/content/migration-and-overseas-workers-2020-census-population-and-housing-2020-cph	Region	5 years	Regional 5 years
povinc	Poverty incidence (2021)	Philippine Statistics Authority (PSA, 2022a) https://psa.gov.ph/content/proportion-poor-filipinos-was-recorded-181-percent-2021	Region	3 years	Regional 3 years
litrate	Literacy rate (2020)	2020 Census of Population and Housing (PSA, 2023a) https://psa.gov.ph/content/literacy-rate-and-educational-attainment-among-persons-five-years-old-and-over-philippines	Region	5 years	Regional 5 years

Table 2
Statistical summary of data inputs.

Scale	Data	Minimum	Maximum	Range	Sum	Median	Mean	SE.mean	Variance	Std.dev.	Count
Provincial Monthly	NTL	0	35,505	35,505	40,798,121	1,226	3,908	6,009	36,111,197	6,009	10,440
	meantemp	19	42	24	297,189	28	28	3	9	3	10,440
	maxrainfall	2	1,779	1,778	3,444,551	288	330	235	55,194	235	10,440
	builtarea	0	263	263	402,394	12	39	55	2,994	55	10,440
Regional monthly	*LI*	0	4	4	842	1	1	1	0	1	612
	caseperH	0	1,828	1,828	43,773	22	72	137	18,865	137	612
Regional ~five-year	*popden*	95	21,510	21,416	957,414	268	1,564	4,994	24,943,296	4,990	612
	popgr	1	3	3	944	1	2	1	0	1	612
	regmov	0	5	5	1,300	2	2	1	1	1	612
	povinc	2	30	28	9,860	15	16	8	57	8	612
	litrate	86	99	13	58,892	97	96	3	8	3	612
	urblev	15	100	85	27,292	38	45	21	462	21	612

performed on the NTL data to remove negative values, unstable light, and clouds as per Del Castillo et al. (2024). After these pre-processing steps, the sum of lights was computed by adding all cloud-corrected pixel values within the region. Fig. 2 shows the provinces with high and low average NTL from January 2013 to December 2022.

3.2.2. Built-up areas

Built-up areas represent impervious surfaces as sources of NTL emission. The built-up area was obtained from MCD12Q1.061 MODIS Land Cover Type Yearly Global of NASA LP DAAC at USGS EROS Center (Friedl and Sulla-Menashe, 2022). The LC_Type1 includes 17 classes. The class of interest was built-up areas, which are spaces with an over 30 % impervious surface. We extracted the built-up class from the data, converted the built-up pixels classified into polygons, and computed the built-up area per province (Fig. 2). The data was downloaded through Google Earth Engine https://developers.google.com/earth-engine/datasets/catalog/MODIS_061_MCD12Q1.

3.2.3. Weather

Previous studies have shown a significant positive correlation

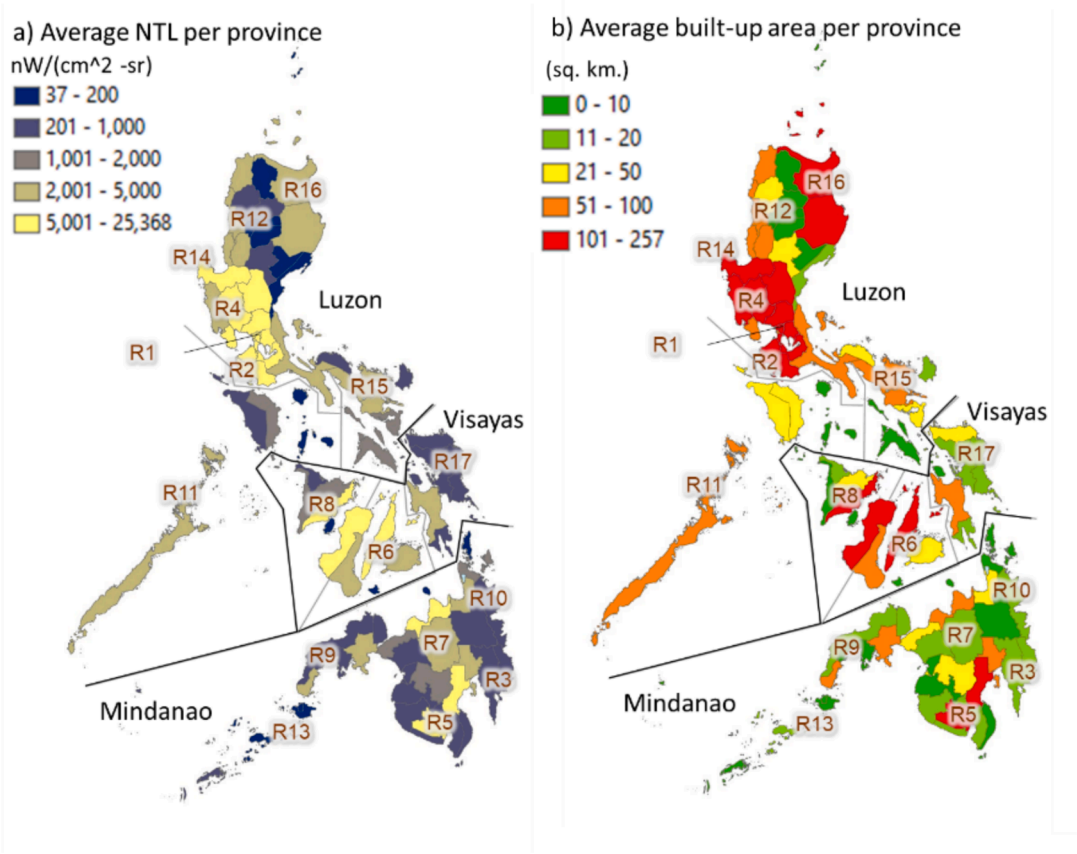


Fig. 2. Provincial data from 2013 to 2022 a) average NTL, b) average built-up area.

between NTL and land surface temperature, attributed to thermal and brightness effects in urban settings (Li et al. 2019, Li et al. 2022). Mean temperature data from MOD11A1.061 Terra Land Surface Temperature and Emissivity Daily Global 1-km dataset were used to estimate NTL. These data were acquired from NASA LP DAAC at the USGS EROS Center (Wan et al. 2021). Specifically, we used the daytime land surface temperature band (LST_Day_1km) with a grid size of 1,200 km. Daily land surface temperature values were averaged monthly for each province.

Conversely, rainfall, indicative of cloud cover, tends to diminish NTL, resulting in a negative relationship. Therefore, including maximum rainfall data in the model indirectly accounts for cloud effects. Additionally, we collected daily rainfall data from the Climate Hazards Group Infrared Precipitation with Station Data Version 2.0 (Funk, et al. 2015), using a precipitation band measured in millimeters per day, with a resolution of 5,566 m. The daily rainfall data were summed up into monthly datasets for each province. Both mean temperature and rainfall datasets were obtained through Google Earth Engine.

3.2.4. Ancillary data

Data on the daily COVID-19 case count (Department of Health 2022b) and the annual count of hospitals (Department of Health 2022a) for each region were obtained from the Department of Health. The daily COVID-19 case data were aggregated from a daily to monthly scale. Additionally, various factors were examined to identify regional characteristics associated with EAR as shown in Fig. 3. Specific regional data were sourced from the PSA for parameters including population growth rate, population density (PSA 2021), urbanization level (PSA 2022b), regional movers (PSA 2023b), poverty incidence (PSA 2022a), and literacy rate (PSA 2023a). Regional movers, also known as "domestic long-distance movers" refer to the population relocating from one province to another within the past five years (PSA 2023b). In this study, the regional mover population was divided by the total regional population. Poverty incidence represents the percentage of the population falling below the poverty line (PSA 2020).

Lockdown levels in the Philippines ranged from Level 1 with the least stringent measures to Level 4 with the strictest restrictions. To analyze them, we computed a Lockdown Index *LI*. This involved recording the number of days *d* in a given month *m* during which each province *p* was under each of these four quarantine levels *l*. We then determined the lockdown duration, *LD*, for each level by multiplying the number of days by the corresponding lockdown weight, *LW*. Lockdown weights were assigned as follows: 0.25 for Level 1, 0.5 for Level 2, 0.75 for Level 3, and 1 for Level 4. Finally, we computed the *LI* for each province and averaged it across each region *r* as shown in Equation (1). We mapped the annual regional COVID-19 cases and the average lockdown index in Fig. 4.

$$LI_{m,r} = \frac{1}{p}\sum_{p=1}^{k}\frac{1}{d}\sum_{d=1}^{n}\left(LD_{d,p}{}^{*}LW_{d,p}\right) \quad (1)$$

3.3. Methods

3.3.1. Framework

Our approach involves developing a model estimating the pre-pandemic NTL. Using this model, we calculated a counterfactual NTL—what the NTL would have been if the pandemic had not occurred. This counterfactual becomes the baseline for calculating the economic impact, represented by the EAR index. We then used regression analysis to identify the regional factors influencing the EAR. The overall process is illustrated in Fig. 5.

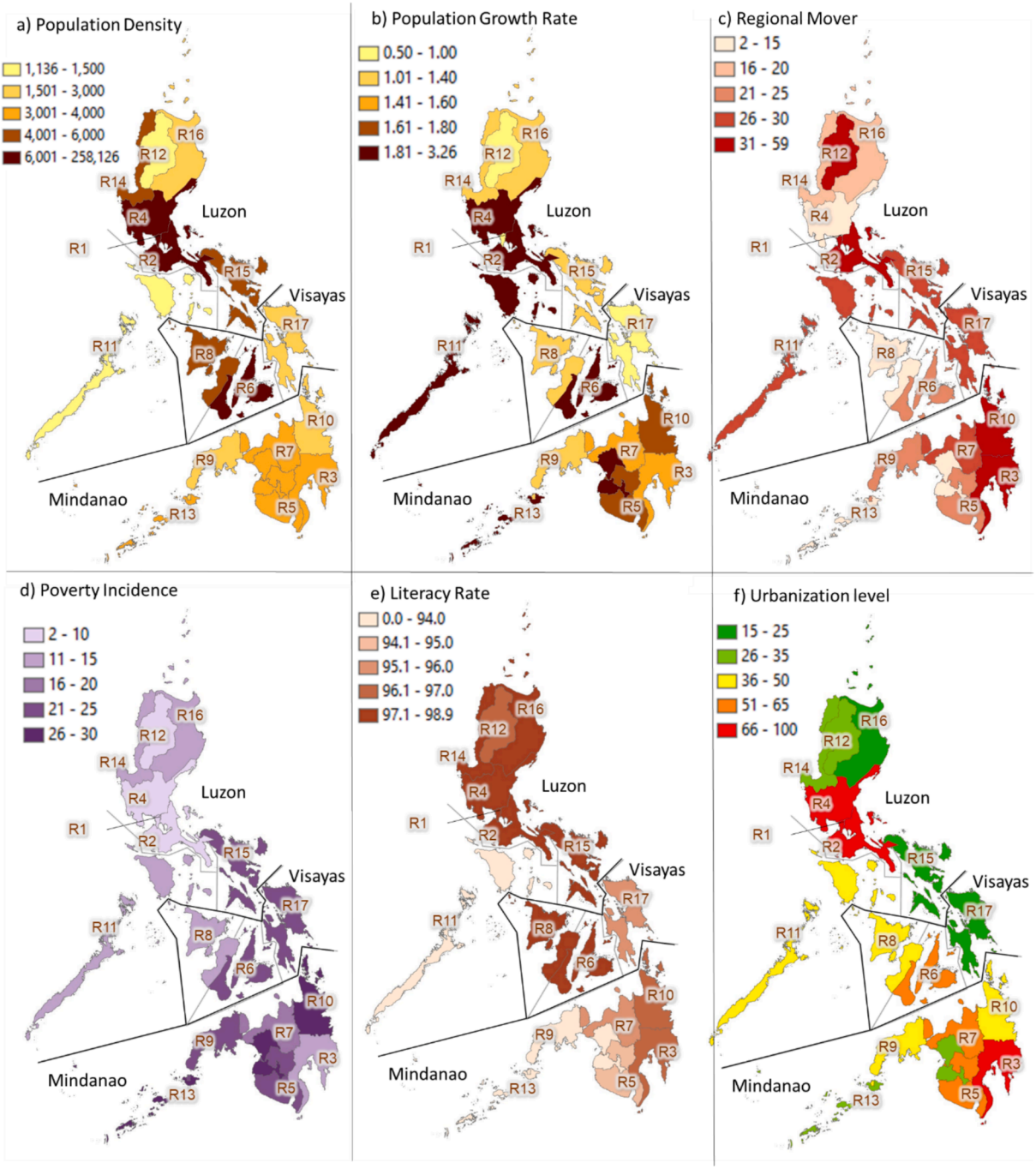


Fig. 3. Regional datasets a) Population density, b) population growth rate, c) regional mover, d) poverty incidence, e) literacy rate, f) urbanization level.

3.3.2. *Data exploration and preparation*

We began by analyzing the distribution of the dataset values. The histograms (Fig. 5) revealed a skewed data distribution. To address this, we applied the natural logarithm to the input datasets to normalize the distribution. Subsequently, we separated the data into the pre-pandemic and pandemic subsets, with March 2020 marking the start of the pandemic.

3.3.3. *NTL model development*

For model development, we first tested the NTL data for spatial autocorrelation using Moran's I to avoid biased estimates. The computed Moran's I was 0.928 (p-value < 0.01), indicating significant positive spatial autocorrelation. This means that spatial regression models had to be used instead of OLS to obtain unbiased results. We then performed Lagrange multiplier (LM) tests to determine which spatial regression model to use. The results showed significant spatial dependence in both the error and the dependent variable so we performed the robust LM. Both had significant p-values but the spatial error model (SEM) had a higher test statistic (Rao score = 25,888; p-value= <0.000) than the spatial lag model (Rao score = 8.36; p-value < 0.004). This means that the residual has greater spatial dependence than the dependent variable.

Based on this, we developed the SEM model where the error term ε of the OLS model was multiplied by the spatial weight matrix W and an autoregressive coefficient λ, thus, isolating an identically and independently distributed error term u. Then, the pre-pandemic NTL, ln*NTL*, was modeled based on the average temperature *lnmeantemp*, maximum rainfall *lnmaxrainfall*, and buil-up area *lnbuiltarea*. Additionally, we

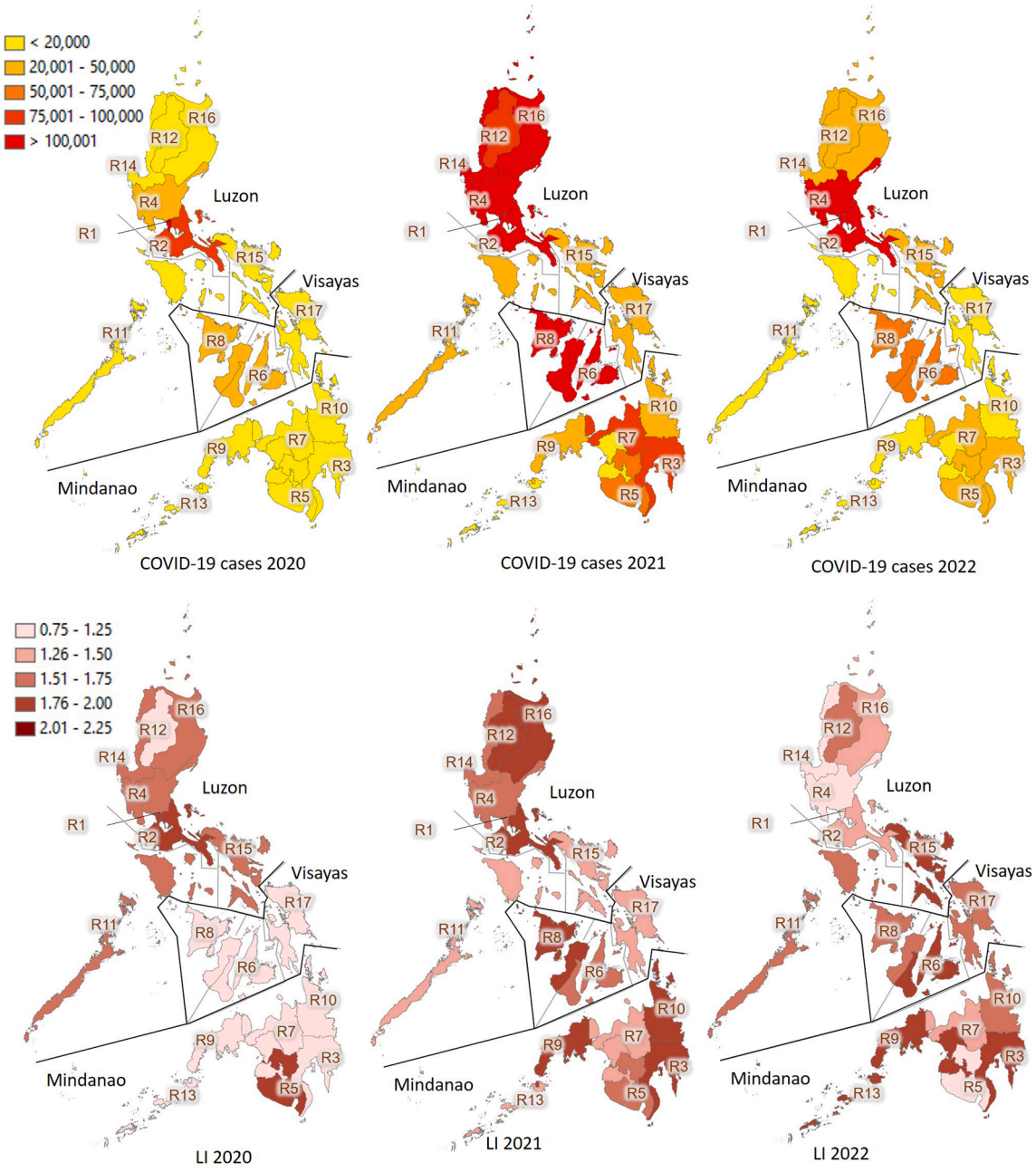


Fig. 4. Total annual regional COVID-19 cases and average annual regional lockdown indices.

incorporated the count of holidays *holidays* to capture the impact of seasonal festivities on NTL. Finally, we added regional dummy variables *R* and *year* component to control for the regional variations and capture the effect of passing years on NTL, respectively, as shown in Equation (2):

$$\ln NTL = \beta_0 + \beta_1 \ln meantemp + \beta_2 \ln maxrainfall + \beta_3 \ln builtarea + \beta_4 holidays + year + R + \lambda W\varepsilon + u \quad (2)$$

where, β_0, β_1, β_2, β_3, and β_4 are the parameters estimated in the model. To determine whether multicollinearity among the explanatory variables exists, we computed the variance inflation factor which falls within an acceptable value (VIF < 3.54).

3.3.4. NTL estimation

We then estimated the provincial monthly NTL by inputting the pandemic period data into the developed SEM. The counterfactual was then estimated using Equation (3).

$$\ln cNTL = \widehat{\beta}_0 + \widehat{\beta}_1 \ln meantemp + \widehat{\beta}_2 \ln maxrainfall + \widehat{\beta}_3 \ln builtarea + \widehat{\beta}_4 holidays + year + R + \lambda W\varepsilon \quad (3)$$

Here, *cNTL* represents the counterfactual NTL. $\widehat{\beta}_0$, $\widehat{\beta}_1$, $\widehat{\beta}_2$, $\widehat{\beta}_3$, and $\widehat{\beta}_4$ are the coefficients obtained from Equation (2).

3.3.5. EAR computation

For the EAR, we aggregated the actual provincial monthly NTL NTL_p into regional monthly NTL NTL_r (Equation (4)):

$$\ln NTL_r = \sum_{p=1}^{k} \ln NTL_p \quad (4)$$

where k stands for the total number of province p in region r.

Similarly, the counterfactual NTL $cNTL_p$ was aggregated into regional level as represented by Equation (5):

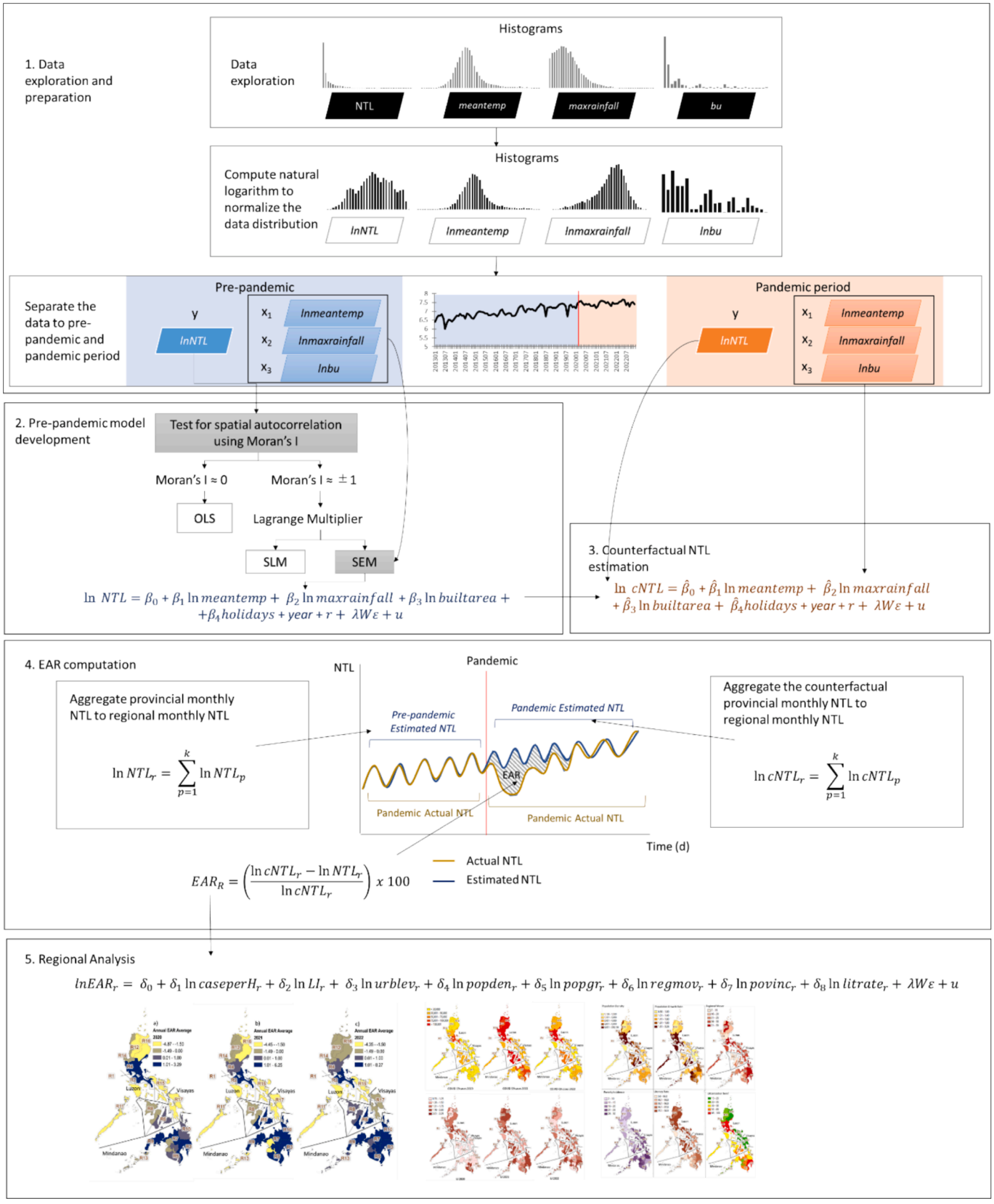


Fig. 5. Overall process flow.

$$\ln cNTL_r = \sum_{p=1}^{k} \ln cNTL_p \quad (5)$$

To compute for the EAR per region *r*, we subtracted the actual NTL from the counterfactual NTL and divided the difference by the counterfactual NTL (Equation (6)).

$$EAR_r = \left(\frac{\ln cNTL_r - \ln NTL_r}{\ln cNTL_r}\right) \quad (6)$$

Here, *EAR* is the ratio of economic activity reduction considered as an index of economic impact.

3.3.6. Regional analysis

We then regressed the EAR on the following factors to determine which factor(s) influence the EAR as represented in Equation (7).

$$\begin{aligned} lnEAR_r = {} & \delta_0 + \delta_1 \ln caseperH_r + \delta_2 \ln LI_r + \delta_3 \ln urblev_r + \delta_4 \ln popden_r \\ & + \delta_5 \ln popgr_r + \delta_6 \ln regmov_r + \delta_7 \ln povinc_r + \delta_8 \ln litrate_r \\ & + \lambda W\varepsilon + u \end{aligned} \quad (7)$$

In this equation, $\ln caseperH_r$ represents the natural logarithm of the ratio of COVID-19 cases per hospital. This ratio was selected because economic impacts are influenced not only by the proportion of the infected population but also by the capacity of healthcare institutions to accommodate patients requiring medical care. $\ln LI_r$ refers to the natural logarithm of the lockdown index (Equation (1)). Other regional factors, such as urbanization level *Urblev*, population density *Popden*, population growth rate *Popgr*, proportion of regional movers *Regmov*, poverty incidence *Povinc*, and literacy rate *Litrate*, could also attenuate or magnify the economic impacts. Previous studies have linked the economic impacts to urbanization level (Carvalho et al. 2021, Hamman et al. 2023) and population density (Khan et al. 2023). Furthermore, regional movers could capture the changes in economic activity owing to the urban exodus that occurred during the lockdown period, especially under the *Balik Probinsya* or Return to Province program (UPPI and DRDF, 2020). Regions with more regional movers, wealth, and higher education levels can also affect the EAR distribution (Carvalho et al. 2021).

4. Results

4.1. Pre-pandemic NTL model

Table 3 presents the results of spatial regression models generated to estimate the pre-pandemic provincial monthly NTL. As shown, we generated three models with varying sets of explanatory variables. Model 1, based on mean temperature, maximum rainfall, built-up areas, and holidays, all variables significant at 1 %, explains 72 % of the pre-pandemic NTL. The year was added in Model 2, effectively increasing the R^2 value to 0.816. Adding the regional dummy variables in Model 3 did not improve the R^2 but slightly lowered the Akaike Information Criterion (AIC), which measures the model quality based on its fit and complexity. Having the lowest AIC, model 3 estimated the counterfactual NTL.

Table 3
SEM results.

	Dependent Variable: *lnNTL*		
	1	2	3
lnmeantemp	1.185***	1.179***	1.177***
	(0.076)	(0.063)	(0.063)
lnmaxrainfall	−0.089***	−0.065***	−0.065***
	(0.005)	(0.004)	(0.004)
lnbuiltarea	1.747***	1.484***	1.589***
	(0.091)	(0.091)	(0.108)
holidays	0.028***	0.005*	0.005*
	(0.003)	(0.003)	(0.003)
R1			0.049
			(0.451)
R2			0.388
			(0.427)
R3			0.657
			(0.429)
R4			−0.328
			(0.396)
R5			0.244
			(0.427)
R6			0.382
			(0.446)
R7			0.608
			(0.434)
R8			0.505
			(0.413)
R9			0.293
			(0.459)
R10			1.012**
			(0.447)
R11			−0.213
			(0.436)
R12			−0.416
			(0.429)
R13			−0.091
			(0.447)
R15			0.258
			(0.414)
R16			−0.764*
			(0.426)
R17			0.381
			(0.421)
year		0.114***	0.114***
		(0.002)	(0.002)
Constant	−2.550***	−230.881***	−231.114***
	(0.407)	(3.861)	(3.876)
Observations	7,482	7,482	7,482
Log Likelihood	−4,166.71	−2,725.27	−2,708.75
R^2	0.719	0.816	0.816
adj-R^2	0.718	0.816	0.816
Akaike Inf. Crit.	8,347.42	5,466.53	5,465.50

Note: Standard errors are shown between parentheses. Significance levels: *p < 0.1; **p < 0.05; ***p < 0.01.
Data coverage: 2013–2019 regional monthly.

4.2. Estimated NTL

We estimated the NTL per province using Model 3. For ease of interpretation, we aggregated the provincial actual and estimated NTL to the regional level. Fig. 6 plots the estimated monthly regional NTL in blue and the actual NTL in black. As shown, the estimated NTL exhibits an ascending trend across the years. It also follows a fluctuating pattern, reflecting seasons controlled by the temperature and rainfall datasets. Focusing on the pre-pandemic period, the estimated NTL generally followed the actual NTL, suggesting that the model can capture the NTL values. However, the modeled values could not capture steep momentary drops possibly due to disasters or power outages and temporary peaks of the actual NTL which occurred in regions surrounding Marawi city during the Marawi Siege.

Regarding the pandemic period, the gaps between the actual and counterfactual NTL represent the economic impacts. During this period, two patterns emerged. The first pattern showed that counterfactual values of R1 to R10 were higher than the actual NTL, indicating economic activity reduction. The second pattern can be observed from R11 to R17 where the actual NTL was higher than the counterfactual, representing economic activity gain.

4.3. EAR

4.3.1. Regional monthly EAR

We computed the regional monthly EAR using Equation (6). EAR values indicate the economic impact based on NTL, as a measure of economic activities. EAR values ranged from −1 to 1 but were

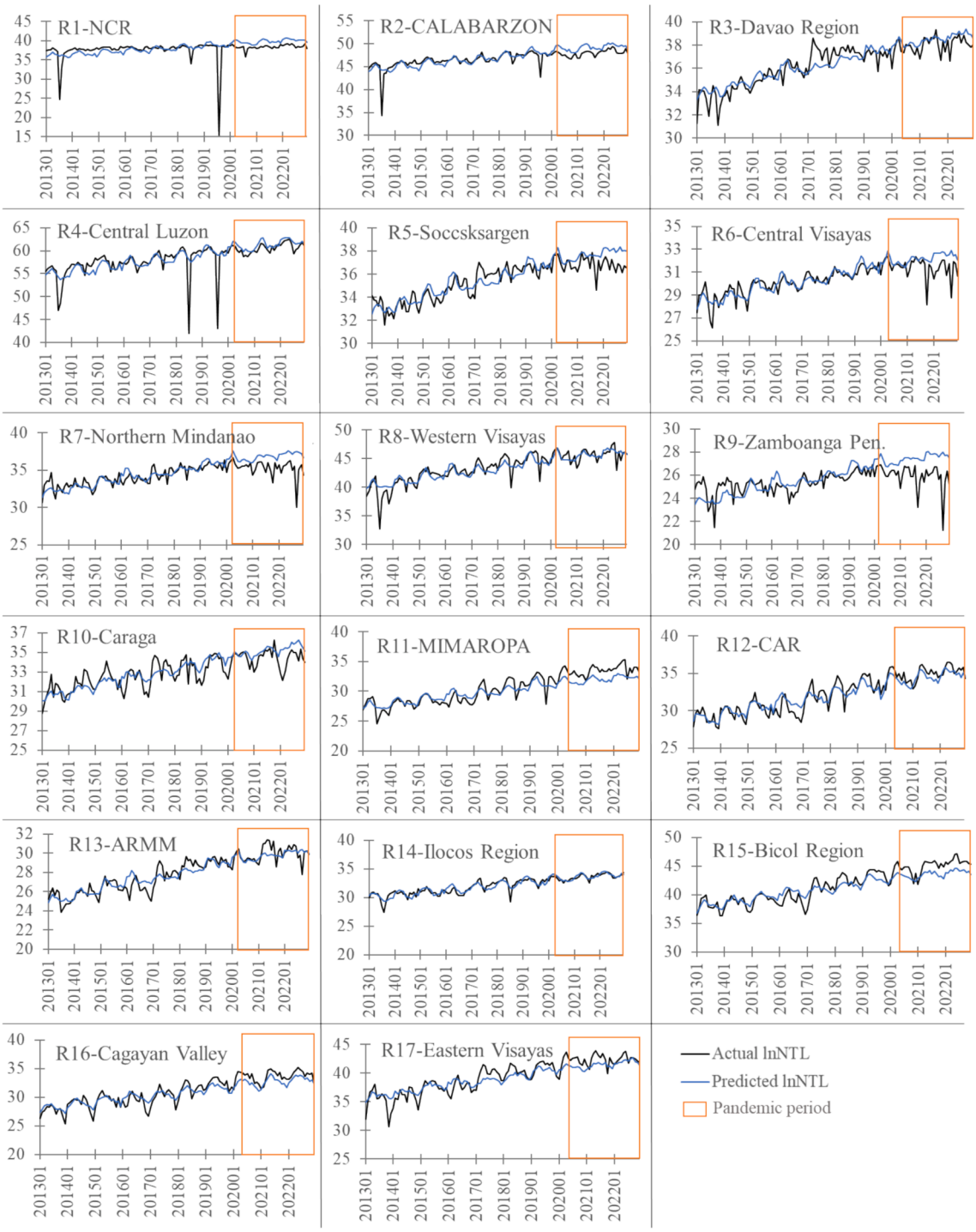


Fig. 6. Monthly actual and estimated NTL values per region.

multiplied by 100 to improve readability. + EAR corresponds to a reduction in economic activities and −EAR reflecting an increase in economic activities. Our computation yielded EAR values from −8.65 to 25.13. Fig. 6 shows the computed EAR values per region and month. The columns represent the region categorized by urbanization ranking, while the rows represent the years and months. EARs were color-coded with yellow for negative and blue for positive values. The graph shows how EAR values within a region change across months. It also shows which regions have the highest EAR for each month. We averaged the EARs for 2020, 2021, and 2022, highlighting regions with worse economic reduction for the year. Then, we averaged the EARs across all months to identify the worst affected regions. We also averaged the EAR per month to determine the months with the greatest economic reduction.

4.3.2. *Spatial distribution of EAR*

We analyzed the spatial distribution of EARs and identified two key patterns. First, the monthly regional EARs showed a clear urban–rural divide. As illustrated in Fig. 7, urban regions (R1–R10) generally experienced + EARs, while rural ones (R11–R17) mostly saw − EARs. This finding indicates that economic impacts varied by region, with urban areas seeing economic reductions, while rural areas, experiencing gains.

Second, the southern regions showed higher EARs. As shown in Fig. 8d, which maps the average EAR per region over three years, Mindanao had significantly more regions with positive EARs—nearly 80 % of the island—compared to only about a third of Luzon and the Visayas. This finding suggests that Mindanao was more severely affected than Luzon and Visayas within the three years.

4.3.3. Temporal distribution of EAR

The three-year-long temporal analysis of EAR values generally showed monthly fluctuations within a region. Specifically, Fig. 7 shows the timing of EAR peaks for each region, each peak occurring at varying months. Two patterns emerged after comparing the timing of regional EAR peaks. The first pattern pertains to the transfer of peaks from regions with high urbanization level to those with lower urbanization level. We highlighted the EAR peaks per region in red boxes (Fig. 7). As shown, the first regional peak occurred in R1 in August 2020. The next peak occurred in R2, then in the less urban regions. Interestingly, this pattern showed that as time passed, the EAR intensified and transferred from the most urbanized regions to less urbanized ones such as R9, R7, and R6.

The second pattern showed a southward movement of the peaks, initially starting in Luzon and moving to Visayas and Mindanao. The peaks in Luzon regions generally occurred within the first year. Contrarily, EAR peaks in Visayas and Mindanao regions occurred in the

	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16	R17	
	NCR	CALB	DAVR	CENL	SOCC	CENV	NORM	WESV	ZAMB	CARA	MIMA	CAR	ARMM	ILOC	BICL	CAGV	EASV	
yyyymr	L	L	M	V	M	V	M	V	M	M	L	L	M	L	L	L	V	Ave
202003	3.59	2.23	-0.20	0.87	0.12	-0.42	2.46	-1.48	2.14	0.33	-4.99	-2.02	-0.71	0.84	-4.17	-3.99	-4.21	-0.57
202004	4.86	2.75	0.82	1.18	1.36	1.71	3.62	-0.47	3.91	0.87	-5.68	-2.32	0.03	0.42	-4.21	-2.49	-4.58	0.10
202005	3.44	2.99	2.54	1.62	1.34	2.37	4.07	-0.92	3.81	0.43	-4.32	-2.19	0.70	0.78	-2.45	-3.00	-3.06	0.48
202006	2.52	2.98	1.35	1.77	0.80	1.99	3.63	-0.05	3.44	0.04	-3.88	0.11	-0.24	0.84	-2.22	-1.04	-2.49	0.56
202007	2.18	2.20	-0.39	0.19	1.59	-0.18	3.13	-2.28	3.53	-0.76	-5.49	-2.50	-0.62	-0.87	-3.87	-3.11	-3.85	-0.65
202008	9.03	2.57	0.49	1.51	-0.57	0.29	3.24	-2.48	3.20	-0.17	-6.48	-1.09	-0.26	0.52	-4.08	-2.02	-3.91	-0.01
202009	3.13	3.88	0.83	1.89	-0.31	0.87	3.23	-4.38	2.33	0.81	-4.49	-3.25	-0.23	-0.50	-4.26	-3.92	-3.94	-0.49
202010	1.74	2.66	2.19	1.51	1.90	0.64	2.24	1.31	1.84	2.30	-3.87	0.02	-0.21	-0.54	-2.46	-0.57	-1.58	0.54
202011	1.26	1.19	2.34	0.20	1.40	0.77	3.41	-0.06	3.75	3.92	-4.61	-1.78	-0.53	-1.72	1.33	-2.46	-2.13	0.37
202012	1.19	0.04	0.35	-0.30	1.41	-1.36	1.01	-2.50	2.46	0.36	-4.85	-2.56	0.80	-1.44	-3.22	-2.01	-5.19	-0.93
202101	2.21	3.15	3.44	0.46	2.93	4.10	5.82	3.95	9.61	4.89	-0.29	2.12	2.38	0.25	2.51	0.66	-2.13	2.71
202102	3.52	4.39	2.37	2.20	2.22	2.97	4.82	3.47	5.34	3.57	-1.47	3.38	0.47	1.03	-0.90	2.75	0.96	2.42
202103	3.53	2.98	-0.80	1.67	-0.56	1.21	3.44	-1.42	2.95	1.00	-4.59	1.43	-3.18	0.77	-3.56	-0.34	-3.64	0.05
202104	5.14	3.24	0.23	1.39	1.72	1.76	3.75	-0.48	3.66	1.32	-5.14	-0.91	-2.85	0.64	-4.04	-2.82	-3.96	0.16
202105	3.58	3.06	0.52	1.71	0.33	-0.32	3.13	-1.91	3.23	1.61	-4.84	-1.46	-5.19	0.06	-4.29	-1.40	-5.55	-0.45
202106	3.33	2.47	-0.58	0.93	-0.19	-0.16	4.41	-1.30	5.76	0.30	-5.58	-1.23	-5.05	-0.65	-3.05	-1.76	-2.92	-0.31
202107	2.13	1.17	0.69	0.25	3.07	1.70	7.36	-0.49	6.43	1.56	-4.68	-1.43	-1.35	-0.53	-4.64	-3.21	-2.42	0.33
202108	4.16	2.51	-1.90	2.54	-1.08	0.80	3.66	-1.42	2.70	0.58	-4.99	-0.88	-4.73	1.70	-4.52	-1.38	-4.53	-0.40
202109	2.01	3.74	1.54	-0.47	2.13	1.63	6.26	-2.70	15.85	3.15	-5.03	-2.67	4.43	-1.87	-4.13	-3.87	-1.23	1.11
202110	3.09	1.75	4.59	0.62	7.87	13.86	11.34	1.86	7.23	-3.25	-5.52	-2.33	-1.32	-1.47	-3.68	-1.17	2.80	2.13
202111	3.90	1.48	0.52	0.37	0.92	1.14	2.76	0.05	4.59	1.18	-5.72	-2.37	-2.40	-0.79	-5.00	-2.09	-3.24	-0.28
202112	0.48	-0.04	2.59	-1.07	3.14	1.57	4.38	-1.09	7.69	2.45	-5.50	-1.32	0.27	-2.41	-4.67	-3.63	-2.26	0.03
202201	3.29	2.24	0.98	1.73	2.66	7.16	4.72	0.52	5.97	7.59	-3.59	-0.27	1.11	0.31	-4.18	-0.54	0.59	1.78
202202	3.79	2.37	5.54	1.07	1.81	5.41	7.90	0.82	8.08	9.34	-3.31	2.21	0.41	0.98	-3.65	-0.42	-0.82	2.44
202203	3.33	1.76	1.70	0.42	2.49	4.71	5.52	0.83	6.53	7.05	-4.19	0.37	-1.38	0.64	-3.54	-1.58	-1.06	1.39
202204	3.61	2.20	0.72	0.45	4.16	3.03	5.20	0.18	6.17	3.50	-5.69	-2.27	-0.30	-0.68	-4.47	-1.43	-1.98	0.73
202205	2.13	1.40	0.76	0.10	2.08	2.48	4.81	-1.83	4.78	2.24	-7.46	-3.34	-1.82	-1.20	-5.75	-3.78	-3.04	-0.44
202206	2.77	1.44	0.50	0.76	4.04	2.44	3.71	-3.75	4.41	2.52	-8.65	-2.38	-1.25	-0.64	-6.11	-1.94	-3.32	-0.32
202207	4.13	3.01	2.00	3.58	4.42	5.90	6.82	6.38	7.72	2.53	1.21	1.48	1.89	1.30	-3.19	-1.55	1.29	2.88
202208	3.63	2.87	0.45	1.48	2.93	2.69	5.71	-1.02	6.86	2.95	-4.98	-0.74	1.91	-0.02	-3.48	-0.10	-0.45	1.22
202209	4.25	3.14	2.79	1.26	3.98	13.85	20.80	3.25	25.13	6.27	-2.63	-1.53	9.00	-0.79	-3.39	-1.45	0.08	4.94
202210	3.43	2.34	2.55	-0.02	4.60	1.54	7.38	-0.72	8.77	1.62	-5.32	-2.64	0.13	-0.99	-4.46	-2.73	-0.94	0.86
202211	1.76	0.85	3.08	0.19	3.24	2.64	5.04	-0.37	5.58	3.69	-4.57	-1.50	0.32	0.26	-3.30	-2.19	-0.65	0.83
202212	3.58	2.04	1.85	0.23	3.54	4.64	7.55	-0.20	9.20	3.80	-3.07	1.74	0.65	-1.47	-4.25	0.94	-1.05	1.75
2020Av	3.29	2.35	1.03	1.04	0.90	0.67	3.00	-1.33	3.04	0.81	-4.87	-1.76	-0.13	-0.17	-2.96	-2.46	-3.49	-0.06
2021Av	3.09	2.49	1.10	0.88	1.88	2.52	5.09	-0.12	6.25	1.53	-4.45	-0.64	-1.54	-0.27	-3.33	-1.52	-2.34	0.62
2022Av	3.31	2.14	1.91	0.94	3.33	4.71	7.10	0.34	8.27	4.42	-4.35	-0.74	0.89	-0.19	-4.15	-1.40	-0.95	1.50
Averag	3.23	2.33	1.37	0.95	2.10	2.75	5.19	-0.31	6.02	2.34	-4.54	-1.00	-0.27	-0.21	-3.51	-1.75	-2.19	0.73

Fig. 7. Regional monthly EAR. L-Luzon, V-Visayas, M-Mindanao. Please refer to Fig. 1 for the regional names and abbreviations. The regions were ranked according to urbanization levels, with R1 having the highest and R17, the lowest. Note: The reported values were multiplied by 100 to improve readability. Red box marks the EAR peaks per region.

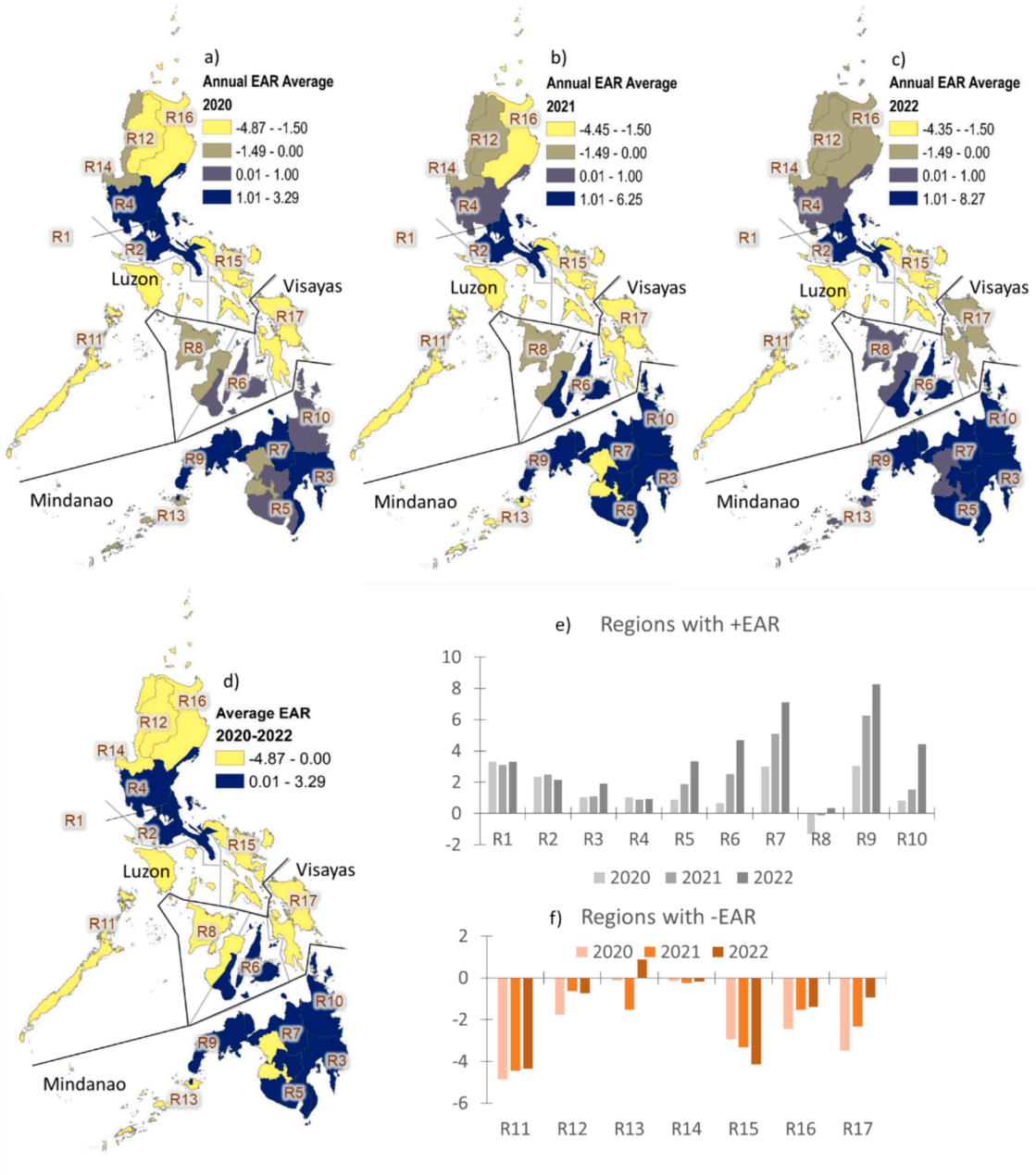


Fig. 8. EAR annual average maps a) 2020, b) 2021, c) 2022 d) 2020 to 2022 and annual EAR trends for e) regions with positive EAR and f) regions with -EAR.

second and third years. Fig. 8a to 8c show the average annual regional EAR for 2020 to 2022. Fig. 8a shows that in 2020, regions with + EAR (blue-colored regions) were in Luzon and Mindanao. However, by 2021, blue regions decreased in Luzon as they expanded and intensified in Mindanao. In 2022, blue regions further increased in Mindanao and Visayas.

Aside from monthly EAR fluctuations, EAR values also reveal recovery trends. Fig. 8e and 8f display the annual average EAR per region for those with positive and negative EAR, respectively. These figures help pinpoint regions showing signs of recovery. Among regions with positive EAR (Fig. 8e), R2 and R4 showed a decline in their annual average EAR, indicating gradual recovery, though none have fully reached their counterfactual values. Contrarily, regions with negative EAR (Fig. 8f), such as R11, R13, R16, and R17, displayed increasing EAR, signaling economic decline despite earlier gains.

4.4. Regional factors

Table 4 presents the result of the regional analysis where *lnEAR* was regressed on several regional attributes to determine the significant factors. We formulated four models. Model 1 covers all data from 2020 to 2022. In Model 1, *lnurblev* and *lnpovinc* were significant at 1 % and 5 %, respectively. The coefficient signs show that *lnurblev* and *lnpovinc* increased the EAR. Although Model 1 indicated insignificance for both *lncaseperH* and *lnLI*, it may have held significance across the years. Investigating this possibility, we divided the data by year and generated Models 2–4. In 2020 and 2021, only *lnurblev* held significance. By 2022, *lnurblev* remained significant while *lncaseperH* became significant at 5 %, which might reflect the emergence of the Omicron variant. At this time, *lnpovinc* was also significant, suggesting that, across time, the economic impacts spread in poorer regions. The significance of *lnurblev* in all four models supports the finding of spatial analysis regarding the separation between urban and rural regions.

5. Discussions

The results revealed that EARs varied across space and time, depending on certain regional factors. EARs were generally higher in urban areas than in rural ones, aligning with the findings of other researchers (Carvalho et al. 2021; Hamman et al. 2023). Our regional analysis confirmed this, with *lnurblev* consistently significant across all models (Table 4). The study also found that while Luzon experienced a substantial initial economic impact following the Luzon lockdowns, the overall medium-term EAR was higher in Mindanao. EARs shifted over time, with peaks moving from the most urbanized regions in the first year to less urbanized ones in the second and third years. EAR peaks also migrated from Luzon to Mindanao. Additionally, the three-year analysis revealed the recovery status of the regions. Key factors linked to EAR included urbanization level, poverty incidence, and number of cases per hospital.

This study demonstrated a counterfactual-based approach to estimating economic impacts, enabling the calculation of medium-term EAR. This method complements the widely-used change detection technique of analyzing NTL datasets. The change detection method measures the difference between pre-and post-event images (Ghosh et al. 2020; Beyer et al. 2021; Alahmadi et al. 2023; Xu et al. 2021) and is appropriate for immediate impact assessment. However, it may underestimate long-term effects, as it does not account for the trend. In contrast, our counterfactual-based approach allows for long-term impact estimation and recovery tracking. Some studies (Lin and Rybnikova, 2023) have used pre-pandemic levels as the recovery point. Alternatively, the counterfactual value can serve as the recovery point in estimating the recovery of losses rather than temporal changes. In the counterfactual approach, recovery is achieved when the actual EAR reaches the counterfactual value (EAR = 0). Although the results show that some regions are moving toward recovery, many still experience positive EAR, and regions with negative EAR show a worsening trend. Hence, continuous monitoring of EAR is crucial for authorities to identify regions in need of further support and to allocate resources effectively for recovery.

Table 4
Regional analysis results.

	lnEAR			
	2020–2022	2020	2021	2022
lncaseperH	0.002	0.002	−0.003	0.010**
	(0.002)	(0.006)	(0.003)	(0.005)
lnLI	0.184	−0.222	−0.475	−0.086
	(0.134)	(0.136)	(0.327)	(0.503)
lnurblev	0.173***	0.127***	0.180***	0.213***
	(0.056)	(0.043)	(0.061)	(0.069)
lnpopden	−0.002	0.001	−0.003	−0.005
	(0.007)	(0.005)	(0.007)	(0.008)
lnpopgr	−0.73	−0.051	−0.904	−1.122
	(1.026)	(0.779)	(1.11)	(1.264)
lnregmov	0.03	0.084	0.068	−0.012
	(0.493)	(0.374)	(0.533)	(0.608)
lnpovinc	0.216**	0.127	0.18	0.333**
	(0.108)	(0.082)	(0.117)	(0.133)
lnlitrate	−0.037	−0.054	0.077	−0.099
	(0.524)	(0.398)	(0.567)	(0.647)
Constant	5.314	4.509	8.549	8.115
	(7.076)	(5.374)	(7.748)	(9.012)
Observations	612	204	204	204
Log Likelihood	1,441.08	528.213	472.143	463.231
sigma2	0.001	0.0003	0.001	0.001
AIC	−2,860.15	−1,034.43	−922.285	−904.462
Wald Test (df = 1)	556.070***	112.771***	144.553***	188.073***
LR Test (df = 1)	276.292***	66.625***	77.816***	97.631***

Note: Standard errors are shown between parentheses. Significance levels: *p < 0.1; **p < 0.05; ***p < 0.01.
Data coverage: 2020–2022 regional annual data.

Medium-term impacts revealed that the most economically stricken regions due to the pandemic were not always the most urbanized regions, which is not covered by studies focusing on the initial lockdown periods (Carvalho et al., 2021; Shinozaki and Rao, 2021). As COVID-19 was a dynamic hazard lasting for several years, the severity of economic impacts fluctuated. This finding implies the need for continuous impact monitoring. Highly urbanized regions often incur more absolute economic loss in the short run than urbanizing regions. Although the absolute losses are lower for urbanizing regions, the impacts to them could be more devastating in the long run as the losses constitute a significant portion of their economy. Neglecting the longer-term impacts could leave these urbanizing regions without support, leading to further decline of their economies, making poorer regions poorer, as implied by poverty incidence having significant positive relationship with EAR in 2022.

We acknowledge several limitations of this study. First, the method requires more data and computation, making it more time-consuming than the simpler change detection approach. Second, the results only reflect economic impacts from NTL-emitting sectors, thus, agricultural impacts are not captured. Third, smaller events, unrelated to the pandemic, may influence the EAR. For example, sudden drops in NTL could be caused by weather disturbances, reduced activity, or power outages. In contrast, the NTL peaks in Mindanao in 2017 might be linked to population displacement during the Marawi Siege. These smaller events were not factored into the pre-pandemic model. Finally, the method assumes no other hazards occurred during the pandemic, though some natural disasters did take place. Nevertheless, these events were short-lived and affected only a few regions, meaning this method is not suited for estimating the impacts of multiple hazards. This implies that EARs were not purely the effect of COVID-19 alone but could also include effects of minor events. Future research could address these limitations by accounting for multiple hazards and integrating multispectral images to capture agricultural losses.

6. Conclusions

The COVID-19 pandemic resulted in global economic activity changes observable from space through NTL data. This study estimated the subnational economic activity reductions at monthly intervals for three years and identified regional factors influencing the changes. We generated a model estimating the pre-pandemic NTL based on explanatory variables including mean temperature, maximum rainfall, built-up area, holidays, trend, and regional dummy variables. The model was able to explain over 81.6 % of the pre-pandemic NTL. Based on the regression model results, we estimated the counterfactual NTL. The pre-pandemic regional estimated NTL closely followed the actual NTL. The regional monthly EARs were then computed. Finally, the regional factors influencing the EARs were identified through regression.

This study's contribution centers on estimating the regional medium-term EAR due to the COVID-19 pandemic using NTL. Our results revealed that EAR was higher in urban regions than in rural regions. EAR was not static over time as shown by the increased impact in less urban and southern regions in the second and third years. The regional analysis showed that urbanization level, COVID-19 cases per hospital, and poverty incidence had a significant positive relationship with EARs. This study showed that impacts differ across space and fluctuate across time depending on regional characteristics. It is important to understand these spatiotemporal dimensions and continuously monitor their long-term impacts. In this regard, NTL holds great potential in understanding these dimensions beyond the immediate impacts.

CRediT authorship contribution statement

Ma. Flordeliza P. Del Castillo: Conceptualization, Data curation, Formal analysis, Methodology, Visualization, Writing – original draft, Writing – review & editing. **Toshio Fujimi:** Conceptualization,

Methodology, Resources, Supervision, Writing – original draft, Writing – review, editing, Fuding acquisition. **Hirokazu Tatano:** Conceptualization, Resources, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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